

# Equilibrium

## Learning & Revision for the Day

- Equilibrium in Physical and Chemical Processes
- Law of Chemical Equilibrium
- Le-Chatelier's Principle
- Ionic Equilibrium
- Degree of Ionisation or Dissociation
- Acids and Bases
- Concept of pH
- Hydrolysis of Salt
- Solubility Product ( $K_{sp}$ )
- Buffer Solutions

In a reversible reaction, the point at which there is no further change in concentration of reactants and products, is called **equilibrium state**. The mixture of reactants and products in the equilibrium state is called an **equilibrium mixture**.

## Equilibrium in Physical and Chemical Processes

It can be established for both physical and chemical processes as discussed below:

- The equilibrium developing between different phases or physical properties of a substance is termed as **physical equilibria**.
  - (i) Solid-liquid equilibrium,  $\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(l)$
  - (ii) Liquid-vapour equilibrium,  $\text{H}_2\text{O}(l) \rightleftharpoons \text{H}_2\text{O}(\text{vap})$
  - (iii) Solid-vapour equilibrium,  $\text{I}_2(s) \rightleftharpoons \text{I}_2(\text{vap})$
- In case of reversible reactions, when the concentration of reactants and products do not change with time, the reaction is called in a state of **chemical equilibrium**.
  - (i) Homogeneous equilibrium,
 
$$\text{H}_2(g) + \text{I}_2(g) \rightleftharpoons 2\text{HI}(g)$$
  - (ii) Heterogeneous equilibrium,
 
$$\text{MgCO}_3(s) \rightleftharpoons \text{MgO}(s) + \text{CO}_2(g) \uparrow$$

## Dynamic Nature of Equilibrium

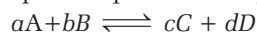
It is always dynamic in nature, i.e. the reaction does not stop but goes on forward and backward directions with equal speed.

e.g.  $\text{solid} \rightleftharpoons \text{liquid}$  (physical equilibria)  
 $\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(l)$  (273 K, 1 atm)

It indicates that at equilibrium, rate of conversion of ice into water = rate of conversion of water into ice

## Law of Chemical Equilibrium

It states that "Rate of a reaction is directly proportional to the concentration of reactants with each concentration term raised to the power equal to the respective stoichiometric coefficient".



Rate of forward reaction  $\propto [A]^a [B]^b = k_f [A]^a [B]^b$

Rate of backward reaction  $\propto [C]^c [D]^d = k_b [C]^c [D]^d$

At equilibrium,  $k_f [A]^a [B]^b = k_b [C]^c [D]^d$ ;

$$\frac{k_f}{k_b} = K_C = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

where,  $K_C$  = equilibrium constant

$k_f$  = rate constant for forward reaction

$k_b$  = rate constant for backward reaction

This expression is also known as law of chemical equilibrium

## Equilibrium Constant

- It is the product of molar concentration of the products, each raised to the power equal to its stoichiometric coefficient divided by the product of the molar concentration of reactants, each raised to the power equal to its stoichiometric coefficient is constant at constant temperature.

- For a gaseous reaction,  $K_p = \frac{P_C^c \cdot P_D^d}{P_A^a \cdot P_B^b}$

where,  $K_p$  = equilibrium constant in terms of partial pressure

- The relation between  $K_p$  and  $K_C$  are given below:

$$K_p = K_C (RT)^{\Delta n_g}$$

$R$  = gas constant,  $T$  = temperature in Kelvin

$\Delta n_g$  = gaseous moles of products

– gaseous moles of reactants

Hence, If  $\Delta n_g = 0$ ,  $K_p = K_C$

If  $\Delta n_g = +ve$ ,  $K_p > K_C$

If  $\Delta n_g = -ve$ ,  $K_p < K_C$

- Equilibrium constant in terms of mole fractions,  $K_x$



$$K_x = \frac{[X_C]^c [X_D]^d}{[X_A]^a [X_B]^b}$$

Similarly,  $K_p = K_x (p)^{\Delta n_g}$

- Equilibrium constant in terms of activities,

$$K_a K_a = \frac{a_C^c \times a_D^d}{a_A^a \times a_B^b}$$

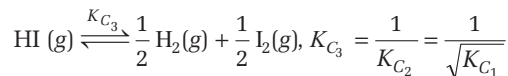
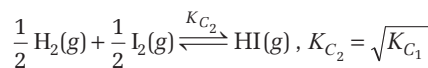
Activity = activity coefficient  $\times$  molality (or molarity)

i.e.  $a = \gamma \times m$

## Some Facts Related to Equilibrium Constant

- Equilibrium constant ( $K_p$  or  $K_C$ ) does not depend on pressure, volume, concentration and catalyst but depends only upon temperature.
- Equilibrium constant for a given reaction is independent of the reaction mechanism.
- Equilibrium constant depends on stoichiometric coefficient

e.g.  $H_2(g) + I_2(g) \xrightleftharpoons{K_{C1}} 2HI(g)$



- If a reaction is multiplied by  $n$ , the rate constant,  $K_C$  becomes  $(K_C)^n$ .  $n$  can be fraction also (+ve only).
- If  $K_1$  be equilibrium constant for  $P \rightleftharpoons Q$  and  $K_2$  be equilibrium constant for  $R \rightleftharpoons S$ , is equilibrium constant for  $P + R \rightleftharpoons Q + S$  is  $K_1 K_2$ .

- Equilibrium constant depends on temperature

For exothermic process  $K_C \propto \frac{1}{T}$

For endothermic process  $K_C \propto T$

## Units of $K_C$ and $K_p$

(i) If  $\Delta n_g = 0$ ,  $K_p = K_C$ , no units for both  $K_C$  and  $K_p$ .

(ii) If  $\Delta n_g > 0$ , unit of  $K_C$  is  $(\text{mol L}^{-1})^{\Delta n_g}$  and that of  $K_p$  is  $(\text{atm})^{\Delta n_g}$ .

(iii) If  $\Delta n_g < 0$ , unit of  $K_C$  is  $(\text{L mol}^{-1})^{\Delta n_g}$  and that of  $K_p$  is  $(\text{atm})^{\Delta n_g}$ .

## Relation between $K$ and $\Delta G^\circ$

Gibbs free energy change and equilibrium constant are related as :

$$\Delta G^\circ = -2.303 RT \log K_C \text{ or } \Delta G^\circ = -2.303 RT \log K_p$$

Significance of  $\Delta G^\circ$  are given below:

(i) If  $\Delta G^\circ < 0$ ,  $\log K < 0 \Rightarrow K < 1$ .  
Therefore, forward reaction is spontaneous.

(ii) If  $\Delta G^\circ > 0$ ,  $\log K < 0 \Rightarrow K < 1$ .  
Therefore, backward reaction is spontaneous.

(iii) If  $\Delta G^\circ = 0$ ,  $\log K = 0 \Rightarrow K = 1$ .  
Therefore, reaction is at equilibrium.

Significance of  $K_C$  and  $K_p$  are given below :

(i) If  $K_C > 10^3$ , products predominate over reactants. In other words, if  $K_C$  is very large, the reaction proceeds almost in all the ways to completion.

(ii) If  $K_C < 10^{-3}$ , reactants predominate over products. In other words, if  $K_C$  is very small, the reaction proceeds hardly at all.

(iii) If  $K_C$  is in the range  $10^{-3}$  to  $10^3$ , appreciable concentration of both reactants and products are present.

## Le-Chatelier's Principle

It states that change in any of the factors that determine the equilibrium conditions of a system, will cause the system to change in such a manner so as to reduce or to counteract the effect of the change.

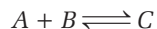
## Factors Affecting Equilibrium

Different factors affecting equilibrium are discussed below:

### 1. Effect of Concentration Change

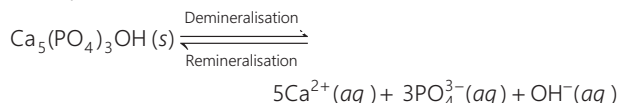
The concentration stress of an added reactant or product is relieved by net reaction in the direction that consumes the added substance.

e.g.



- If we increase the concentration of either  $A$  or  $B$  (reactants), the equilibrium goes in the direction that consumes  $A$  or  $B$ , i.e. forward side.
- If we increase the concentration of  $C$  (product), the equilibrium goes in the direction that consumes  $C$ , i.e. backward side.
- If we remove  $C$  (product), the equilibrium goes in the direction in which its concentration increases, i.e. forward side.
- If any of the species is in solid or in liquid state, its addition does not alter the original equilibrium.

**NOTE** Sweet substances cause tooth decay because on fermentation these produce  $H^+$  ions which combine with  $OH^-$  ions and shift the equilibrium in forward direction



### 2. Effect of Pressure

At high pressure, reaction goes from higher moles to lower moles or from higher volume to lower volume and *vice versa*.

- If  $\Delta n_g = 0$ , no effect on equilibrium due to pressure change.
- If  $\Delta n_g > 0$ , the increase in pressure favours backward reaction.
- If  $\Delta n_g < 0$ , the increase in pressure favours forward reaction. ( $\Delta n_g$  = number of moles of gaseous products – number of moles of gaseous reactants).
- Flash evaporation technique is used for concentrating some aqueous solutions which cannot be concentrated by normal boiling. Concentration of this type of solution is carried out under reduced pressure below  $100^\circ C$ .

### 3. Effect of Temperature

At high temperature, reaction goes to endothermic direction while at low temperature reaction goes to exothermic direction. The equilibrium constant for an endothermic reaction (positive  $\Delta H$ ) increases as the temperature increases.

$$K \propto T; \text{ if } \Delta H^\circ = +ve \text{ (endothermic)}$$

$$K \propto \frac{1}{T}; \text{ if } \Delta H^\circ = -ve \text{ (exothermic)}$$

Freeze drying technique is used for drying heat sensitive substances. In this technique, water is made to sublime off at a temperature below  $0^\circ C$ .

### 4. Effect of Catalyst

A catalyst increases the rate of forward reaction as well as the rate of backward reaction, so it does not affect the equilibrium and equilibrium constant.

### 5. Effect of Inert Gas

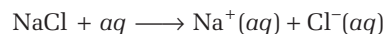
At constant volume, there is no effect of addition of inert gas. At constant pressure, when inert gas is added, reaction goes from lower moles to higher moles.

## Ionic Equilibrium

Existence of ions in equilibrium with its electrolytes is called ionic equilibrium. The species furnish ions can be a weak or strong electrolyte.

## Weak and Strong Electrolytes

- Separation of an electrolyte into their ions either on fusion or dissolution is called **ionisation** or **dissociation**.



- **Weak electrolytes** is defined as a substance which dissociates partially into their ions in the solutions and such solutions are weak conductor of electricity. e.g.  $CH_3COOH$ ,  $H_3PO_4$ ,  $H_3BO_3$ ,  $NH_4OH$ ,  $HCN$ , etc.
- **Strong electrolytes** is defined as a substance which dissociates completely into their ions in solution and such solutions are very good conductor of electricity, e.g.  $HCl$ ,  $H_2SO_4$ ,  $NaOH$ ,  $KOH$ ,  $NaCl$ ,  $KCl$ , etc.
- Ionic equilibrium can be defined as equilibrium which is established between the unionised molecules and the ions in the solution of weak electrolytes.
- The weak electrolytes are only partially ionised and a dynamic equilibrium is maintained after sometime between the ions and unionised molecules.
- **Ionisation of acids and bases** Acids and bases dissociate into the ions on ionisation. The acids which completely dissociates into their constituent ions in an aqueous medium are called as **strong acids**, e.g.  $HCl$ ,  $HNO_3$ ,  $H_2SO_4$ , etc. while the bases which are completely dissociated into ions in an aqueous medium are called **strong bases**, e.g.  $NaOH$ ,  $Ba(OH)_2$  etc.

(Usually the term dissociation is used for weak electrolyte and ionisation for strong electrolyte).

## Degree of Ionisation or Dissociation

- It is the fraction of total number of moles undergoing ionisation and is denoted by  $(\alpha)$ .

Alternately, the fraction of the amount of an electrolyte present in the solution as free ions is called degree of ionisation  $(\alpha)$ .

$$\alpha = \frac{\text{number of moles of electrolyte dissociated as ions}}{\text{total number of moles of electrolyte dissolved}}$$

$$\text{or } \alpha = \frac{\text{amount of electrolyte dissociated}}{\text{initial amount}}$$

- **Degree of dissociation** ( $\alpha$ ) of a gaseous compound is related to its vapour density by  $\alpha = \frac{D-d}{d(y-1)}$

Here,  $D$  = molar density before dissociation/initial density

$d$  = density after dissociation/density of the gaseous mixture

$y$  = number of moles of products

$$\text{Density of gas} = \frac{\text{Molecular weight}}{2}$$

$$\text{Molecular weight} = \text{Density of gas} \times 2$$

$$M_O = D \times 2 \text{ or } M_C = d \times 2$$

Here,  $M_O$  = Observed molecular weight (Abnormal)

$M_C$  = Calculated molecular weight (Theoretical)

## Ostwald's Dilution Law

- It states that degree of dissociation of weak electrolyte is inversely proportional to the square root of concentration.

$$\alpha \propto \frac{1}{\sqrt{C}}, \quad \alpha = \sqrt{\frac{K}{C}} = \sqrt{KV}$$

where,  $\alpha$  = degree of dissociation

$V$  = volume containing 1 mole of weak electrolyte

$K$  = dissociation constant.

- This law is used to calculate the degree of dissociation,  $\alpha$  for weak acids and bases from the known value of  $K$ .

## Limitations of Ostwald's Law

Ostwald's law has following limitations :

- This law holds good only in case of weak electrolytes. Even weak electrolytes do not obey this law in concentrated solutions.
- The value of  $K$  can be calculated only in dilute solutions of weak electrolytes.

## Factors Influencing Degree of Ionisation/Dissociation

Ionisation or dissociation is influenced by the following factors:

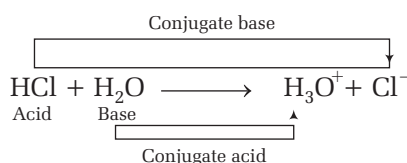
- For strong electrolyte,  $\alpha = 1$  at normal dilution, while for most of the polar covalent compounds, i.e. weak electrolytes,  $\alpha \ll 1$ .
- Degree of ionisation of an electrolyte increases with polarity of the solvent.
- The degree of ionisation of an electrolyte decreases with increase in concentration of the electrolyte.
- The degree of ionisation rises with raise in temperature of the solution.
- The addition of species possessing a common ion to that of weak electrolyte causes a decrease in the degree of dissociation of weak electrolyte.

## Acids and Bases

An acid is that whose aqueous solution tastes sour, turns blue litmus red, neutralises bases and so on. On the other hand, the aqueous solution of a base tastes bitter, turns red litmus blue, neutralises acid and so on.

Various concepts of acids and bases are given below:

- According to **Arrhenius concept**, the substances which release  $H^+$  ions on dissolving in water are called acids and which release  $OH^-$  ions on dissolving in water are called bases.
- According to **Bronsted-Lowry concept**, acids are proton donors, while bases are proton acceptors.



- According to **Lewis concept**, acids are the substances which accept a pair of electrons to form a coordinate bond, e.g.,  $BF_3$ ,  $AlCl_3$ , etc. and bases are the substances which donate a pair of electrons to form coordinate bond. e.g.,  $NH_3$ ,  $CN^-$ ,  $OH^-$ ,  $I^-$  etc.
- (i) **Acid strength** The extent to which the acidic property is shown by an acid is a measure of its strength.
- (ii) The strength of an acidic solution does not depend upon its concentration but depend upon the number of  $H^+$  ions present.
- (iii) To measure the relative strength of the two acids, the measurements of their  $H^+$  ions concentration, i.e. the degree of dissociation of their equinormal solutions is used.

$$\frac{\text{Strength of acid I}}{\text{Strength of acid II}} = \frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_1/C}{K_2/C}} = \sqrt{\frac{K_1}{K_2}}$$

where,  $K_1$  and  $K_2$  are the dissociation constants of two acids at the same concentration  $C$ .

$\alpha_1$  and  $\alpha_2$  are their degree of dissociation.

## Ionisation of Polybasic Acids

- The acids which have more than one ionisable proton per molecule of acids are known as polybasic or polyprotic acids, e.g. oxalic acid ( $HOOC-COOH$ ),  $H_2SO_4$  and  $H_3PO_4$ .
- The ionisation reactions for a dibasic acid,  $H_2X$  are represented as



The corresponding equilibrium constants are as:

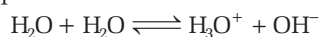
$$K_{a_1} = \frac{[H^+][HX^-]}{[H_2X]} \text{ and } K_{a_2} = \frac{[H^+][X^{2-}]}{[HX^-]}$$

( $K_{a_1}$  and  $K_{a_2}$  are called first and second ionisation constants, respectively, of the acid  $H_2X$ ).

- Similarly, for tribasic acids such as  $\text{H}_3\text{PO}_4$ , there are three ionisation constants.
- It is found that  $K_{a_1}$  of polyprotic acid is greater than  $K_{a_2}$  or  $K_{a_3}$  (i.e.  $K_{a_1} > K_{a_2} > K_{a_3}$ ), because it is more difficult to remove a positively charged proton from a negative ion due to electrostatic forces.
- Similarly, it is more difficult to remove a proton from a doubly charged anion as compared to single charged anion. The primary reaction involves the dissociation of  $\text{H}_2\text{X}$  and  $\text{H}_3\text{O}^+$  in the solution comes mainly from the first dissociation step.

## Ionic Product of Water ( $K_w$ )

- Pure water is a weak electrolyte and is ionised according to following equation.



At 25°C, for pure water

$$[\text{H}_3\text{O}^+] = [\text{OH}^-] = 10^{-7} \text{ mol/L}$$

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14} \text{ and } K = \frac{K_w}{55.55}$$

where,  $K_w$  = ionic product of water,

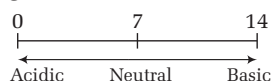
$K$  = ionisation constant.

- Value of  $K_w$  depends upon temperature. If temperature increases, value of  $K_w$  also increases.

## Concept of pH

It is used to express and compare the acidic and basic strength of a solution.

- It is defined as the negative logarithm of  $\text{H}_3\text{O}^+$  ion concentration (in moles per litre) present in it.  
Thus,  $\text{pH} = -\log [\text{H}_3\text{O}^+]$   
Similarly,  $\text{pOH} = -\log [\text{OH}^-] \Rightarrow \text{pH} + \text{pOH} = 14$
- Hydronium ion concentration in molarity is more conveniently expressed on a logarithmic scale known as **pH scale**.
- **pH scale range** is 0 to 14 and it depends upon the value of  $K_w$ . As temperature increases, value of pH decreases at 25°C. pH scale range will be



- pH of very dilute ( $\sim 10^{-8}$  M or lower) acids or bases is nearly 7 but not 7 (i.e. not simply  $-\log [\text{acid or base}]$ ) due to ionisation of water.
- pH of strong acids with concentration  $> 1$  M is never negative, it is zero only.

## pH of Mixtures of Acids and Bases

The rules for determining the pH of mixtures of acids and bases are as follows:

- If strong acid or strong base remains unused, calculate the concentration or molarity of  $\text{H}^+$  ions and  $\text{OH}^-$  ions left in the solution and then calculate the pH or pOH accordingly.

- If weak acid or weak base is left behind or remains unused, a buffer (acidic or basic) is formed. Calculate the concentration of salt formed (mmoles of salt formed/volume of solution) and the concentration of weak acid or weak base left behind. Use the buffer equation to calculate the pH of the solution.
- If acids or bases are completely neutralised, then salt is formed. Calculate the concentration of the salt formed and use the hydrolysis equation to calculate the pH of the solution.

## Hydrolysis of Salt

- The process of salt hydrolysis is actually the reverse process of neutralisation.
- The reaction of an anion or cation of the salt with water accompanied by cleavage of O—H bond, is called **hydrolysis of salt**.
- Salt hydrolysis affects the pH of the solution. On the basis of relative strengths of acid and base, the salts are divided into four categories:
  - Salts of strong acids and strong bases (i.e. neutral salts) do not undergo hydrolysis, e.g.  $\text{NaCl}$ ,  $\text{CaSO}_4$  etc. If such salt is dissolved in water, pH of the solution remains 7.
  - Salt of a strong acid and weak base, e.g.  $\text{NH}_4\text{Cl}$  are called **acidic salts**. Such salts undergo cationic hydrolysis. pH of acidic salt solution will be less than 7.

For such salts,  $[\text{H}_3\text{O}^+] = \sqrt{K_h \times C}$

$$K_h = \frac{K_w}{K_b} \text{ or } \text{pH} = 7 - \frac{1}{2} [\text{p}K_b + \log C]$$

where,  $K_h$  = hydrolysis constant

$K_b$  = ionisation constant for weak base

$C$  = molar concentration of salt.

- Salt of strong base and weak acid, e.g.  $\text{NaNO}_2$ ,  $\text{NaCN}$ ,  $\text{CH}_3\text{COONa}$  are termed as **basic salts**. Such salts undergo anionic hydrolysis.

pH of basic salt solution will be more than 7.

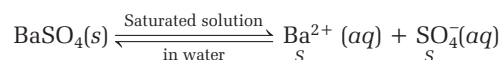
For basic salts,  $[\text{OH}^-] = \sqrt{K_h \times C}$  or  $K_h = \frac{K_w}{K_a}$

$$\text{pH} = 7 + \frac{1}{2} [\text{p}K_a + \log C]$$

- The salts other than halides, sulphates, nitrates of metals fall into this category. e.g.  $\text{CH}_3\text{COONH}_4$  etc.

For such salts,

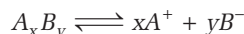
$$K_h = \frac{K_w}{K_a \times K_b}, \text{pH} = 7 + \frac{1}{2} [\text{p}K_a - \text{p}K_b]$$



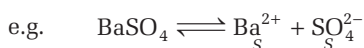
There is no effect of dilution on the hydrolysis of salts of weak acid and weak base because  $h$ , pH and  $K_h$  are all independent of concentration,  $C$ .

## Solubility Product ( $K_{sp}$ )

It is defined as the product of molar concentration of its ions in a saturated solution, each concentration terms raised to the power equal to the number of ions produced on dissociation of one molecule of electrolyte.



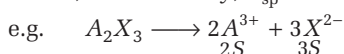
$$\therefore K_{sp} = [A^+]^x [B^-]^y$$



$$K_{sp} = [Ba^{2+}][SO_4^{2-}]$$

$$\therefore K_{sp} = S^2 \text{ or } S = \sqrt{K_{sp}}$$

where,  $S$  = solubility,  $K_{sp}$  = solubility product



$$K_{sp} = [A^{3+}]^2 [X^{2-}]^3$$

$$\Rightarrow K_{sp} = (2S)^2 (3S)^3$$

$$\therefore K_{sp} = 108S^5$$

## Ionic Product (Q)

(At any stage of reaction)

It is the product of ions and give the direction of reaction.

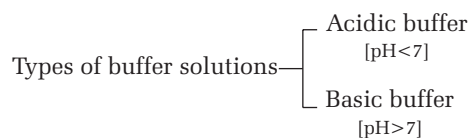
If  $Q > K_{sp}$ , precipitate will be formed.

If  $Q < K_{sp}$ , precipitate does not form.

If  $Q = K_{sp}$ , reaction is at equilibrium.

## Buffer Solutions

It is the solution, which maintains its pH constant or reserve acidic or basic nature even upon addition of small amounts of acid or base. The ability of buffer solution to resist changes in pH on addition of acid or base is called buffer action.



## Henderson's Equation

- A buffer solution pH of which is less than 7 is called **acidic buffer**. Weak acid + salt of the acid with strong base  
e.g.  $CH_3COOH + CH_3COONa$ ;  $HCN + NaCN$
- A buffer solution having pH more than 7 is called **basic buffer**. Weak base + salt of the base with strong acid  
e.g.  $NH_4OH + NH_4Cl$ ,  $C_6H_5NH_2 + C_6H_5NH_3^+Cl^-$
- A buffer solution should exhibit following characteristics:
  - Buffer solutions possess a definite pH value.
  - Their pH value remains constant on keeping long or on dilution.
  - The pH value is not changed on the addition of a strong acid in acidic buffer and a strong base in basic buffer.
- For acidic buffer;  $pH = pK_a + \log \frac{[salt]}{[acid]}$
- For basic buffer;  $pOH = pK_b + \log \frac{[salt]}{[base]}$

$$pH + pOH = 14$$

## Buffer Capacity

The quantitatively defined as the number of moles of acid or base added to 1 L in of butter solution to change the pH by unity.

Buffer capacity

$$= \frac{\text{number of moles of acid / base added to 1 L of buffer}}{\text{change in pH}}$$

Buffer capacity is maximum when

- $[salt] = [acid]$ ,  $pH = pK_a$  for acidic buffer
- $[salt] = [base]$ ,  $pH = pK_b$  for basic buffer

Greater the buffer capacity, larger is its capacity to resist the change in pH value.

## Common-Ion Effect

- It can be defined as a shift in equilibrium on adding a substance that provides more of an ionic species already present in the dissociation equilibrium.  
e.g.  $NH_4OH \rightleftharpoons NH_4^+ + OH^-$
- If  $NH_4Cl$  or  $NaOH$  is added to  $NH_4OH$  solution, the above equilibrium will shift to the left due to high concentration of common ion and therefore, the ionisation of  $NH_4OH$  is further suppressed.



## DAY PRACTICE SESSION 1

## FOUNDATION QUESTIONS EXERCISE

- 1 4 moles of A are mixed with 4 moles of B. At equilibrium for the reaction,  $A + B \rightleftharpoons C + D$ , 2 moles of C and D are formed. The equilibrium constant for the reaction will be  
(a) 1 (b) 1/2 (c) 4 (d) 1/4
- 2 If the value of an equilibrium constant for a particular reaction is  $1.6 \times 10^{12}$  then at equilibrium the system will contain  
→ CBSE-AIPMT 2015  
(a) all reactants  
(b) mostly reactants  
(c) mostly products  
(d) similar amounts of reactants and products
- 3 The following equilibria are given  
I.  $N_2 + 3H_2 \rightleftharpoons 2NH_3$ ;  $K_1$   
II.  $N_2 + O_2 \rightleftharpoons 2NO$ ;  $K_2$   
III.  $H_2 + \frac{1}{2} O_2 \rightleftharpoons H_2O$ ;  $K_3$   
The equilibrium constant for the reaction,  
 $2NH_3 + \frac{5}{2} O_2 \rightleftharpoons 2NO + 3H_2O$   
in terms of  $K_1$ ,  $K_2$  and  $K_3$  will be  
→ NEET 2017  
(a)  $K_1 K_3^2 / K_2$  (b)  $K_2 K_3^3 / K_1$  (c)  $\frac{K_2 K_3}{K_1}$  (d)  $\frac{K_2^3 K_3}{K_1}$
- 4 One mole of a compound AB reacts with one mole of a compound CD according to the equation  $AB + CD \rightleftharpoons AD + CB$ . When equilibrium had been established it was found that 3/4 mole each of reactants AB and CD had been converted to AD and CB. There is no change in volume. The equilibrium constant for the reaction is  
(a)  $\frac{9}{16}$  (b)  $\frac{1}{9}$  (c)  $\frac{16}{19}$  (d) 9
- 5 A mixture of 0.3 moles of  $H_2$  and 0.3 mole of  $I_2$  is allowed to react in a 10 L evacuated flask at  $500^\circ C$ . The reaction is  $H_2 + I_2 \rightleftharpoons 2HI$ , the K is found to be 64. The amount of unreacted  $I_2$  at equilibrium is  
(a) 0.03 mol (b) 0.06 mol  
(c) 0.09 mol (d) 3.6 mol
- 6 For the reaction,  
 $H_2(g) + CO_2(g) \rightleftharpoons CO(g) + H_2O(g)$ ;  
If initial concentration of  $[H_2] = [CO_2]$  and x mol/L of hydrogen is consumed at equilibrium, the correct expression of  $K_p$  is  
(a)  $\frac{x^2}{(1-x)^2}$  (b)  $\frac{x^2}{(2+x)^2}$  (c)  $\frac{x^2}{1-x^3}$  (d)  $\frac{(1+x)^2}{(1-x)^2}$
- 7 At 3000 K, the equilibrium pressure of  $CO_2$ , CO and  $O_2$  are 0.6, 0.4 and 0.2 atm respectively.  $K_p$  for the reaction,  $2CO_2 \rightleftharpoons 2CO + O_2$  is  
(a) 0.089 (b) 0.098  
(c) 0.189 (d) 0.198
- 8 In which of the following equilibrium  $K_c$  and  $K_p$  are not equal?  
→ CBSE-AIPMT 2010  
(a)  $2C(s) + O_2(g) \rightleftharpoons 2CO_2(g)$   
(b)  $SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$   
(c)  $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$   
(d)  $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$
- 9 For the reaction,  
 $CH_4(g) + 2O_2(g) \rightleftharpoons CO_2(g) + 2H_2O(l)$ ;  
 $\Delta_r H = -170.8 \text{ kJ mol}^{-1}$   
Which of the following statements is not true?  
(a) At equilibrium, the concentration of  $CO_2(g)$  and  $H_2O(l)$  are not equal  
(b) The equilibrium constant for the reaction is given by  
 $K_p = \frac{[CO_2]}{[CH_4][O_2]}$   
(c) Addition of  $CH_4(g)$  or  $O_2(g)$  at equilibrium will cause a shift to the right  
(d) The reaction is exothermic
- 10  $K_p = 0.04$  atm at 899 K for the equilibrium shown below. The equilibrium concentration of  $C_2H_6$  when it placed in a flask at 4.0 atm pressure and allowed to come to equilibrium is  
 $C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$   
(a) 0.30 (b) 3.6 (c) 4.0 (d) 2.8
- 11 The value of  $K_p$  for the following reaction  
 $2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$ ; is  $1.2 \times 10^{-2}$  at  $106.5^\circ C$ . The value of  $K_c$  for this reaction is  
(a)  $1.2 \times 10^{-2}$  (b)  $< 1.2 \times 10^{-2}$   
(c)  $> 1.2 \times 10^{-2}$  (d) None of these
- 12 Which of the following statement is correct for a reversible process in a state of equilibrium?  
→ CBSE-AIPMT 2015  
(a)  $\Delta G = -2.30RT \log K$  (b)  $\Delta G = 2.30RT \log K$   
(c)  $\Delta G^\circ = -2.30RT \log K$  (d)  $\Delta G^\circ = 2.30RT \log K$
- 13 What is the effect of increasing pressure on the dissociation of  $PCl_5$ , according to the equation?  
 $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) - x \text{ cal}$   
(a) Dissociation decreases  
(b) Dissociation increases  
(c) Dissociation does not change  
(d) None of the above

- 14** Of the following which change will shift the equilibrium towards the product?



- (a) Increase in temperature  
(b) Increase in total pressure  
(c) Increase in concentration of I  
(d) Decrease in concentration of  $\text{I}_2$
- 15** For the reaction,  $\text{PCl}_3(g) + \text{Cl}_2(g) \rightleftharpoons \text{PCl}_5(g)$ , the position of equilibrium can be shifted to the right by
- (a) doubling the volume  
(b) increasing the temperature  
(c) addition of  $\text{Cl}_2$  at constant volume  
(d) addition of equimolar quantities of  $\text{PCl}_3$  and  $\text{PCl}_5$

- 16** For the reversible reaction,



The equilibrium shifts in forward direction

→ CBSE-AIPMT 2014

- (a) By increasing the concentration of  $\text{NH}_3(g)$   
(b) By decreasing the pressure  
(c) By decreasing the concentration of  $\text{N}_2(g)$  and  $\text{H}_2(g)$   
(d) By increasing the pressure and decreasing temperature
- 17** The value of  $\Delta H$  for the reaction,  
 $\text{X}_2(g) + 4\text{Y}_2(g) \rightleftharpoons 2\text{XY}_4(g)$  is less than zero. Formation of  $\text{XY}_4(g)$  will be favoured at
- CBSE-AIPMT 2011
- (a) low pressure and low temperature  
(b) high temperature and low pressure  
(c) high pressure and low temperature  
(d) high temperature and high pressure

- 18** The degree of dissociation of 1.0 M weak acid, HA is 0.5%. If 2 mL of 1.0 M HA solution is diluted to 32 mL, the degree of dissociation of the acid and  $\text{H}_3\text{O}^+$  ion concentration in the resulting solution will be respectively?

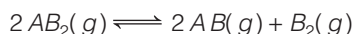
- (a) 0.02 and  $8.2 \times 10^{-12}$  (b) 0.02 and  $3.125 \times 10^{-4}$   
(c) 0.02 and  $1.25 \times 10^{-3}$  (d)  $1.25 \times 10^{-3}$  and 0.02

- 19** The percentage of pyridine ( $\text{C}_5\text{H}_5\text{N}$ ) that forms pyridinium ion ( $\text{C}_5\text{H}_5\text{N}^+\text{H}$ ) in a 0.10M aqueous pyridine solution ( $K_b$  for  $\text{C}_5\text{H}_5\text{N} = 1.7 \times 10^{-9}$ ) is

→ NEET 2016, Phase II

- (a) 0.0060% (b) 0.013%  
(c) 0.77% (d) 1.6%

- 20** The dissociation equilibrium of a gas  $\text{AB}_2$  can be represented as



The degree of dissociation is  $x$  and is small compared to 1. The expression relating the degree of dissociation ( $x$ ) with equilibrium constant  $K_p$  and total pressure  $p$  is

→ CBSE-AIPMT 2008

- (a)  $(2K_p/p)$  (b)  $(2K_p/p)^{1/3}$   
(c)  $(2K_p/p)^{1/2}$  (d)  $(K_p/p)$

- 21** The conjugate base of  $\text{H}_2\text{PO}_4^-$  is

- (a)  $\text{PO}_4^{3-}$  (b)  $\text{P}_2\text{O}_5$  (c)  $\text{H}_3\text{PO}_4$  (d)  $\text{HPO}_4^{2-}$

- 22** Which of the following is least likely to behave as Lewis base?

→ CBSE-AIPMT 2011

- (a)  $\text{NH}_3$  (b)  $\text{BF}_3$  (c)  $\text{OH}^-$  (d)  $\text{H}_2\text{O}$

- 23** Ammonium ion is

- (a) neither an acid nor a base  
(b) both an acid and a base  
(c) a conjugate acid  
(d) a conjugate base

- 24** Which of the following fluoro-compounds is most likely to behave as a Lewis base?

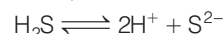
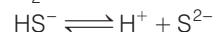
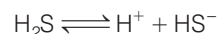
→ NEET 2016, Phase II

- (a)  $\text{BF}_3$  (b)  $\text{PF}_3$  (c)  $\text{CF}_4$  (d)  $\text{SiF}_4$

- 25** Which of the following is electron deficient?

- (a)  $(\text{CH}_3)_2$  (b)  $(\text{SiH}_3)_2$  (c)  $(\text{BH}_3)_2$  (d)  $\text{PH}_3$

- 26**  $K_{a1}$ ,  $K_{a2}$  and  $K_{a3}$  are the respective ionisation constant for the following reactions.



The correct relationship between  $K_{a1}$ ,  $K_{a2}$  and  $K_{a3}$  is

- (a)  $K_{a3} = K_{a1} \times K_{a2}$  (b)  $K_{a3} = K_{a1} \times K_{a2}$   
(c)  $K_{a3} = K_{a1} - K_{a2}$  (d)  $K_{a3} = K_{a1} / K_{a2}$

- 27** What is the pH of 0.01 M glycine solution? For glycine  $K_{a1} = 4.5 \times 10^{-3}$  and  $K_{a2} = 1.7 \times 10^{-10}$  at 298 K?

- (a) 3.0 (b) 10.0 (c) 6.1 (d) 7.1

- 28** Given pH of a solution A is 3 and it is mixed with another solution B having pH 2. If both mixed then resultant pH of the solution will be

- (a) 3.2 (b) 1.9 (c) 3.4 (d) 3.5

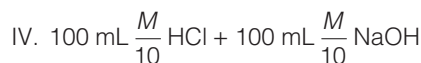
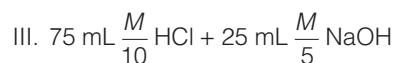
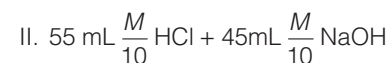
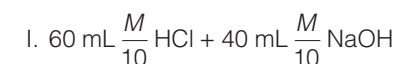
- 29** Hydrogen ion concentration in mol/L in a solution of pH = 5.4 will be

- (a)  $3.98 \times 10^{-6}$  (b)  $3.68 \times 10^{-6}$   
(c)  $3.88 \times 10^{-6}$  (d)  $3.98 \times 10^{-8}$

- 30** pH of 0.08 mol  $\text{dm}^{-3}$  HOCl solution is 2.85. Calculate its ionisation constant.

- (a)  $24.9 \times 10^{-6}$  (b)  $28.6 \times 10^{-5}$   
(c)  $2.5 \times 10^{-6}$  (d)  $24.9 \times 10^{-5}$

- 31** Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations:



→ NEET-2018

pH of which one of them will be equal to 1?

- (a) IV (b) I (c) II (d) III



- 32** By adding 20 mL of 0.1 N HCl to 20 mL 0.001 N KOH, the pH of the obtained solution will be  
(a) 2 (b) 1.3 (c) 0 (d) 7
- 33** Equimolar solutions of the following substances were prepared separately. Which one of these will record the highest pH value? **→ CBSE-AIPMT 2012**  
(a)  $\text{BaCl}_2$  (b)  $\text{AlCl}_3$  (c)  $\text{LiCl}$  (d)  $\text{BeCl}_2$
- 34** The ionisation constant of ammonium hydroxide is  $1.77 \times 10^{-5}$  at 298 K. Hydrolysis constant of ammonium chloride is **→ CBSE-AIPMT 2009**  
(a)  $5.65 \times 10^{-10}$  (b)  $6.50 \times 10^{-12}$   
(c)  $5.65 \times 10^{-13}$  (d)  $5.65 \times 10^{-12}$
- 35** Concentration of the  $\text{Ag}^+$  ions in a saturated solution of  $\text{Ag}_2\text{C}_2\text{O}_4$  is  $2.2 \times 10^{-4} \text{ mol l}^{-1}$  solubility product of  $\text{Ag}_2\text{C}_2\text{O}_4$  is **→ NEET 2017**  
(a)  $2.42 \times 10^{-8}$  (b)  $2.66 \times 10^{-12}$   
(c)  $4.5 \times 10^{-11}$  (d)  $5.3 \times 10^{-12}$
- 36** The solubility of  $\text{BaSO}_4$  in water is  $2.42 \times 10^{-3} \text{ g L}^{-1}$  at 298 K. The value of its solubility product ( $K_{\text{sp}}$ ) will be (Given molar mass of  $\text{BaSO}_4 = 233 \text{ g mol}^{-1}$ ) **→ NEET 2018**  
(a)  $1.08 \times 10^{-14} \text{ mol}^2 \text{L}^{-2}$  (b)  $1.08 \times 10^{-12} \text{ mol}^2 \text{L}^{-2}$   
(c)  $1.08 \times 10^{-10} \text{ mol}^2 \text{L}^{-2}$  (d)  $1.08 \times 10^{-8} \text{ mol}^2 \text{L}^{-2}$
- 37**  $\text{MY}$  and  $\text{NY}_3$ , two nearly insoluble salts, have the same  $K_{\text{sp}}$  values of  $6.2 \times 10^{-13}$  at room temperature. Which statement would be true in regard to  $\text{MY}$  and  $\text{NY}_3$ ? **→ NEET 2016, Phase I**  
(a) The molar solubility of  $\text{MY}$  in water is less than that of  $\text{NY}_3$   
(b) The salts  $\text{MY}$  and  $\text{NY}_3$  are more soluble in 0.5M  $\text{KY}$  than in pure water  
(c) The addition of the salt of  $\text{KY}$  to solution of  $\text{MY}$  and  $\text{NY}_3$  will have no effect on their solubilities  
(d) The molar solubilities of  $\text{MY}$  and  $\text{NY}_3$  in water are identical.
- 38** The solubility of  $\text{AgCl(s)}$  with solubility product  $1.6 \times 10^{-10}$  in 0.1 M  $\text{NaCl}$  solution would be **→ NEET 2016, Phase II**  
(a)  $1.26 \times 10^{-5} \text{ M}$  (b)  $1.6 \times 10^{-9} \text{ M}$   
(c)  $1.6 \times 10^{-11} \text{ M}$  (d) zero
- 39** If pH of a saturated solution of  $\text{Ba(OH)}_2$  is 12, the value of its  $K_{\text{sp}}$  is **→ CBSE-AIPMT 2010**  
(a)  $4.00 \times 10^{-6} \text{ M}^3$  (b)  $4.00 \times 10^{-7} \text{ M}^3$   
(c)  $5.00 \times 10^{-7} \text{ M}^3$  (d)  $5.00 \times 10^{-6} \text{ M}^3$
- 40** In which of the following solvents is silver chloride most soluble?  
(a)  $0.1 \text{ mol dm}^{-3} \text{ AgNO}_3$  solution  
(b)  $0.1 \text{ mol dm}^{-3} \text{ HCl}$  solution  
(c)  $\text{H}_2\text{O}$   
(d) Aqueous ammonia
- 41** A buffer solution is used in  
(a) preparation of potash alum  
(b) the removal of  $\text{PO}_4^{3-}$  ions  
(c) increasing the pH value of a solution  
(d) precipitation of  $\text{Cr(OH)}_3$  from  $\text{CrCl}_3$
- 42** An aqueous solution of 1 M  $\text{NaCl}$  and 1 M  $\text{HCl}$  is  
(a) not a buffer but  $\text{pH} < 7$  (b) not a buffer but  $\text{pH} = 7$   
(c) a buffer with  $\text{pH} < 7$  (d) a buffer with  $\text{pH} > 7$
- 43** 0.1 mole of  $\text{CH}_3\text{NH}_2$  ( $K_b = 5 \times 10^{-4}$ ) is mixed with 0.08 moles of  $\text{HCl}$  and diluted to one litre. What will be the  $\text{H}^+$  concentration in the solution?  
(a)  $8 \times 10^{-2} \text{ M}$  (b)  $8 \times 10^{-11} \text{ M}$   
(c)  $1.6 \times 10^{-11} \text{ M}$  (d)  $8 \times 10^{-5} \text{ M}$
- 44** Which one of the following pairs of solution is not an acidic buffer? **→ CBSE-AIPMT 2015**  
(a)  $\text{HClO}_4$  and  $\text{NaClO}_4$   
(b)  $\text{CH}_3\text{COOH}$  and  $\text{CH}_3\text{COONa}$   
(c)  $\text{H}_2\text{CO}_3$  and  $\text{Na}_2\text{CO}_3$   
(d)  $\text{H}_3\text{PO}_4$  and  $\text{Na}_3\text{PO}_4$
- 45** A physician wishes to prepare a buffer solution at  $\text{pH} = 3.58$  that efficiently resist changes in pH yet contains only small concentration of the buffering agents. Which one of the following weak acid together with its sodium salt would be best to use?  
(a) *m*-chlorobenzoic acid ( $\text{p}K_a = 3.98$ )  
(b) *p*-chlorocinnamic acid ( $\text{p}K_a = 4.41$ )  
(c) 2,5-dihydroxy benzoic acid ( $\text{p}K_a = 2.97$ )  
(d) Acetoacetic acid ( $\text{p}K_a = 3.58$ )
- 46** Buffer solutions have constant acidity and alkalinity because **→ CBSE-AIPMT 2012**  
(a) these give unionised acid or base on reaction with added acid or alkali  
(b) acids and alkalies in these solutions are shielded from attack by other ions  
(c) they have large excess of  $\text{H}^+$  or  $\text{OH}^-$  ions  
(d) they have fixed value of pH
- 47** A buffer solution is prepared in which the concentration of  $\text{NH}_3$  is 0.30 M and the concentration of  $\text{NH}_4^+$  is 0.20 M. If the equilibrium constant,  $K_b$  for  $\text{NH}_3$  is equal  $1.8 \times 10^{-5}$ , what is the pH of this solution? ( $\log 2.7 = 0.43$ ) **→ CBSE-AIPMT 2011**  
(a) 9.43 (b) 11.72  
(c) 8.73 (d) 9.08
- 48** What is  $[\text{H}^+]$  in mol/L of a solution that is 0.20 M in  $\text{CH}_3\text{COONa}$  and 0.10 M in  $\text{CH}_3\text{COOH}$ ? ( $K_a$  for  $\text{CH}_3\text{COOH} = 1.8 \times 10^{-5}$ ) **→ CBSE-AIPMT 2010**  
(a)  $3.5 \times 10^{-4}$  (b)  $1.1 \times 10^{-5}$   
(c)  $1.8 \times 10^{-5}$  (d)  $9.0 \times 10^{-6}$
- 49** In a buffer solution containing equal concentration of  $\text{B}^-$  and  $\text{HB}$ , the  $K_b$  for  $\text{B}^-$  is  $10^{-10}$ . The pH of buffer solution is **→ CBSE-AIPMT 2010**  
(a) 10 (b) 7  
(c) 6 (d) 4
- 50** In which of the following, dissociations of  $\text{NH}_4\text{OH}$  will be minimum? **→ CBSE-AIPMT 2010**  
(a)  $\text{NaOH}$  (b)  $\text{H}_2\text{O}$   
(c)  $\text{NH}_4\text{Cl}$  (d)  $\text{NaCl}$

## DAY PRACTICE SESSION 2

## PROGRESSIVE QUESTIONS EXERCISE

- 1** A buffer solution with pH 9 is to be prepared by mixing  $\text{NH}_4\text{Cl}$  and  $\text{NH}_4\text{OH}$ . Calculate the number of moles of  $\text{NH}_4\text{Cl}$  that should be added to one litre of 1.0 M  $\text{NH}_4\text{OH}$ .  $[K_b = 1.8 \times 10^{-5}]$
- (a) 3.4 (b) 2.6  
(c) 1.5 (d) 1.99
- 2** For a reaction,  
(i)  $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$ ,  $p_{\text{CO}_2} = 2 \text{ atm}$   
(ii)  $\text{CO}_2(l) + \text{C}(s) \rightarrow 2\text{CO}(g)$ ;  $K = 6 \text{ atm}$   
Hence, equilibrium constant for the reaction  
 $\text{CaCO}_3(s) + \text{C}(s) \rightleftharpoons \text{CaO}(s) + 2\text{CO}(g)$  is
- (a) 8 atm (b) 4 atm  
(c) 12 atm (d) 3 atm
- 3** At constant temperature, the equilibrium constant ( $K_p$ ) for the decomposition reaction.  $\text{N}_2\text{O}_4(g) \rightleftharpoons 2\text{NO}_2(g)$  is expressed by  $K_p = \frac{(4x^2p)}{(1-x^2)}$ ; where,  $p$  = pressure,  
 $x$  = extent of decomposition, which one of the following statements is true?
- (a)  $K_p$  increases with increase of  $p$   
(b)  $K_p$  remains constant with change in  $p$  and  $x$   
(c)  $K_p$  increases with increase of  $x$   
(d) None of the above
- 4** An amount of solid  $\text{NH}_4\text{HS}$  is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. Ammonium hydrogen sulphide decomposes to yield  $\text{NH}_3$  and  $\text{H}_2\text{S}$  gases in the flask. When the decomposition reaction reaches at equilibrium, the total pressure in the flask rises to 0.84 atm. The equilibrium constant for  $\text{NH}_4\text{HS}$  decomposition at this temperature is
- (a) 0.11 (b) 0.22 (c) 0.33 (d) 0.44
- 5** pH of a buffer solution decreases by 0.02 units when 0.12 g of acetic acid is added to 250 mL of a buffer solution of acetic acid and potassium acetate at  $27^\circ\text{C}$ . The buffer capacity of the solution is
- (a) 0.1 (b) 10 (c) 1 (d) 0.4
- 6** At a given temperature, equilibrium is attained when 50% of each reactant is converted into the products  
 $\text{A}(g) + \text{B}(g) \rightleftharpoons \text{C}(g) + \text{D}(g)$   
If amount (in mol) of  $\text{B}(g)$  is doubled, percentage of  $\text{B}$  converted into products will be
- (a) 100% (b) 50%  
(c) 66.67% (d) 33.33%
- 7** Which of the following, on the addition will cause deep red colour to disappear?
- $$\text{Fe}^{3+}(aq) + \text{SCN}^- \rightleftharpoons \text{Fe}(\text{SCN})^{2+}(aq)$$
- (Pale yellow) (Colourless) (Deep red)
- I:  $\text{AgNO}_3$ ; II:  $\text{HgCl}_2$   
III:  $\text{H}_2\text{C}_2\text{O}_4$  (oxalic acid)
- (a) I, II (b) II, III  
(c) I, III (d) I, II, III
- 8** The rapid change of pH near the stoichiometric point of an acid-base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentration of the conjugate acid ( $\text{HIn}$ ) and base ( $\text{In}^-$ ) forms of the indicator given by the expression
- (a)  $\log \frac{[\text{In}^-]}{[\text{HIn}]} = \text{p}K_{\text{In}} - \text{pH}$   
(b)  $\log \frac{[\text{HIn}]}{[\text{In}^-]} = \text{p}K_{\text{In}} - \text{pH}$   
(c)  $\log \frac{[\text{HIn}]}{[\text{In}^-]} = \text{pH} - \text{p}K_{\text{In}}$   
(d)  $\log \frac{[\text{In}^-]}{[\text{HIn}]} = \text{pH} - \text{p}K_{\text{In}}$
- 9** The ionisation constant of an acid,  $K_a$ , is the measure of strength of an acid. The  $K_a$  values of acetic acid, hypochlorous acid and formic acid are  $1.74 \times 10^{-5}$ ,  $3.0 \times 10^{-8}$  and  $1.8 \times 10^{-4}$  respectively. Which of the following orders of pH of 0.1 mol  $\text{dm}^{-3}$  solutions of these acids is correct?
- (a) Acetic acid > hypochlorous acid > formic acid  
(b) Hypochlorous acid > acetic acid > formic acid  
(c) Formic acid > hypochlorous acid > acetic acid  
(d) Formic acid > acetic acid > hypochlorous acid
- 10** For the following equilibrium,  $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$  in gaseous phase,  $\text{NO}_2$  is 50% of the total volume when equilibrium is setup. Hence, per cent of dissociation of  $\text{N}_2\text{O}_4$  is
- (a) 50% (b) 25% (c) 66.66% (d) 33.33%
- 11** For the following reaction,  
 $2\text{SO}_2 + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g)$   
The equilibria relation between  $K_C$  and  $K_x$  is
- (a)  $K_C = K_x (RT)^{-1}$  (b)  $K_C = K_x (RT)$   
(c)  $K_C = K_x \left(\frac{1}{V}\right)$  (d)  $K_C = K_x (V)$
- 12** A weak acid  $\text{HX}$  has the dissociation constant  $1 \times 10^{-5}$  M. It forms a salt  $\text{NaX}$  on reaction with alkali. The degree of hydrolysis of 0.1 M solution of  $\text{NaX}$  is
- (a) 0.0001% (b) 0.01% (c) 0.1% (d) 0.15%

**13** For the following equilibrium: (omitting charges)



then relation between  $K, \beta_1, \beta_2$  and  $\beta_3$  is

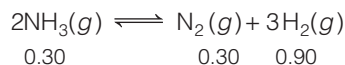
(a)  $p_K = p_{\beta_1} + p_{\beta_2} + p_{\beta_3}$

(b)  $K = \beta_1 \beta_2 \beta_3$

(c)  $\log K = \log \beta_1 + \log \beta_2 + \log \beta_3$

(d) All of the above

**14** When  $NH_3$  is heated in a 0.50 L flask at 700 K and 100 atm pressure, it decomposes into  $N_2$  and  $H_2$  and their equilibrium moles are given below :



Which of the pairs given represents correct value?

Initial moles of  $NH_3$      $K_C$  for  $NH_3$  formation

(a) 0.80                      9.72

(b) 0.90                      9.72

(c) 0.80                      0.103

(d) 0.90                      0.103

**15** If  $x$  moles of lead acetate and 0.1 mole of acetic acid in 1 L solution make a solution of  $pH = 5.04$ . Then, find the value of  $x$ , If  $pK_a$  of  $CH_3COOH$  is 4.74

(a) 0.2 mol                      (b) 0.05 mol

(c) 0.1 mol                      (d) 0.02 mol

**16** A solution which is  $10^{-3}$  M each in  $Mn^{2+}, Fe^{2+}, Zn^{2+}$  and  $Hg^{2+}$  is treated with  $10^{-16}$  M sulphide ion. If  $K_{sp}$  of  $MnS, FeS, ZnS$  and  $HgS$  are  $10^{-15}, 10^{-23}, 10^{-20}$  and  $10^{-54}$  respectively, which one will precipitate first?

(a) FeS                      (b) MgS

(c) HgS                      (d) ZnS

**17**  $HCOOH$  and  $CH_3COOH$  solution have equal pH. If  $K_1/K_2$  (ratio of acid ionisation constant) is 4, their molar concentration ratio will be

(a) 2                      (b) 0.5                      (c) 4                      (d) 0.25

**18**  $pK_b$  of  $NH_3 A^-$  is 4.74 and  $pK_b$  of  $B^-$  and  $C^-$  are 4, 5 and 6, respectively. Aqueous solution of 0.01 M has pH in the increasing order

(a)  $NH_4 A < NH_4 B < NH_4 C$

(b)  $NH_4 C < NH_4 B < NH_4 A$

(c)  $NH_4 C < NH_4 A < NH_4 B = 7$

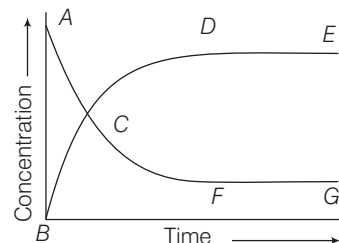
(d) All have equal pH being salt of weak acid and weak base

**19**  $N_2O_4 \rightleftharpoons 2NO_2, K_C = 4$ . This reversible reaction is studied graphically as shown in the given figure. Select the correct statements out of I, II and III.

I. Reaction quotient has maximum value at point A.

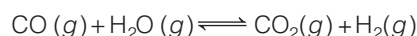
II. Reaction proceeds left to right at a point when  $[N_2O_4] = [NO_2] = 0.1$  M

III.  $K_C = Q$  when point D or F is reached.



(a) I, II                      (b) II, III                      (c) I, III                      (d) I, II, III

**20** 1 mole each of  $CO(g), H_2O(g), H_2(g)$ , and  $CO_2(g)$  are place in 1 L flask at  $25^\circ C$ . When following equilibrium is setup



$K_p = 9$ . Hence,  $CO_2$  present at equilibrium is

(a) 0.5 mol                      (b) 1.5 mol                      (c) 0.25 mol                      (d) 3.0 mol

## ANSWERS

### SESSION 1

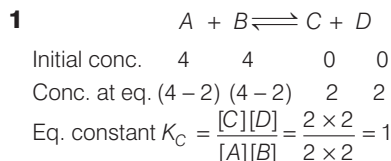
|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1 (a)  | 2 (c)  | 3 (b)  | 4 (d)  | 5 (b)  | 6 (a)  | 7 (a)  | 8 (a)  | 9 (b)  | 10 (b) |
| 11 (b) | 12 (c) | 13 (a) | 14 (a) | 15 (c) | 16 (d) | 17 (c) | 18 (c) | 19 (b) | 20 (b) |
| 21 (d) | 22 (b) | 23 (c) | 24 (b) | 25 (c) | 26 (b) | 27 (d) | 28 (b) | 29 (a) | 30 (a) |
| 31 (d) | 32 (b) | 33 (a) | 34 (a) | 35 (d) | 36 (c) | 37 (a) | 38 (b) | 39 (c) | 40 (d) |
| 41 (b) | 42 (a) | 43 (b) | 44 (a) | 45 (d) | 46 (a) | 47 (a) | 48 (d) | 49 (d) | 50 (c) |

### SESSION 2

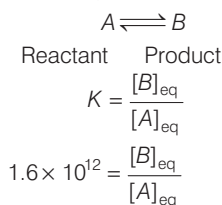
|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1 (d)  | 2 (c)  | 3 (b)  | 4 (a)  | 5 (d)  | 6 (d)  | 7 (d)  | 8 (d)  | 9 (b)  | 10 (d) |
| 11 (d) | 12 (b) | 13 (d) | 14 (d) | 15 (c) | 16 (c) | 17 (d) | 18 (b) | 19 (b) | 20 (b) |

# Hints and Explanations

## SESSION 1



**2** For a reaction,



$$\therefore [B]_{\text{eq}} \gg [A]_{\text{eq}}$$

So, mostly the product will be present in the equilibrium mixture.

**3** From given equations,

$$\text{We have, } K_1 = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

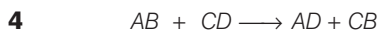
$$K_2 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$$

$$K_3 = \frac{[\text{H}_2\text{O}]}{[\text{H}_2][\text{O}_2]^{1/2}}$$

For the given reaction,

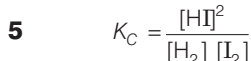
$$K = \frac{[\text{NO}]^2 [\text{H}_2\text{O}]^3}{[\text{NH}_3]^2 [\text{O}_2]^{5/2}}$$

$$\frac{K_2 K_3^3}{K_1} = \frac{[\text{NO}]^2 [\text{H}_2\text{O}]^3}{[\text{NH}_3]^2 [\text{O}_2]^{5/2}}, K = \frac{K_2 K_3^3}{K_1}$$



|                |                                |                                |               |               |
|----------------|--------------------------------|--------------------------------|---------------|---------------|
| Initial conc.  | 1                              | 1                              | 0             | 0             |
| Conc. at equi. | $\left(1 - \frac{3}{4}\right)$ | $\left(1 - \frac{3}{4}\right)$ | $\frac{3}{4}$ | $\frac{3}{4}$ |

$$K_C = \frac{[AD][CB]}{[AB][CD]} = \frac{\left\{\frac{3}{4}\right\} \left\{\frac{3}{4}\right\}}{\left\{\frac{1}{4}\right\} \left\{\frac{1}{4}\right\}} = 9$$



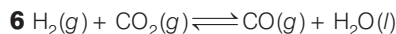
$$64 = \frac{x^2}{0.03 \times 0.03}$$

$$x^2 = 64 \times 9 \times 10^{-4}$$

$$x = 8 \times 3 \times 10^{-2} = 0.24$$

$x$  is the amount of HI at equilibrium,

$$\therefore \text{Amount of } \text{I}_2 \text{ at equilibrium} \\ = 0.30 - 0.24 = 0.06$$



|                |         |         |     |     |
|----------------|---------|---------|-----|-----|
| Initial conc.  | 1       | 1       | 0   | 0   |
| Conc. at equi. | $(1-x)$ | $(1-x)$ | $x$ | $x$ |

$$K_p = \frac{p_{\text{CO}} \cdot p_{\text{H}_2\text{O}}}{p_{\text{H}_2} \cdot p_{\text{CO}_2}} = \frac{x \cdot x}{(1-x)(1-x)} \\ = \frac{x^2}{(1-x)^2}$$

**7.**  $K_p = \frac{[p_{\text{CO}}]^2 [p_{\text{O}_2}]}{[p_{\text{CO}_2}]}$

$$= \frac{[0.4]^2 \times [0.2]}{[0.6]^2} = 0.0888 \approx 0.089$$

**8** The reaction for which the number of moles of gaseous products ( $n_p$ ) is not equal to the number of moles of gaseous reactants ( $n_R$ ), has different value of  $K_C$  and  $K_p$ .

- (a)  $n_p \neq n_R$ , thus,  $K_p \neq K_C$   
 (b)  $n_p = n_R = 2$ , thus,  $K_p = K_C$   
 (c)  $n_p = n_R = 2$ , thus,  $K_p = K_C$   
 (d)  $n_p = 2, n_R = 2$  thus,  $K_p = K_C$

**9** This is an example of heterogeneous chemical equilibrium

$$K_C = \frac{[\text{CO}_2]}{[\text{CH}_4][\text{O}_2]^2}$$

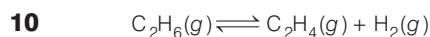
and  $K_p = \frac{p_{\text{CO}_2}}{p_{\text{CH}_4} \times p_{\text{O}_2}^2}$

(according to partial pressure)

Thus, concentration of  $\text{CO}_2(\text{g})$  and  $\text{H}_2\text{O}(\text{l})$  are not equal to equilibrium.

$$K_p = \frac{[\text{CO}_2]}{[\text{CH}_4][\text{O}_2]}$$

expression.



|               |                 |     |     |
|---------------|-----------------|-----|-----|
| Initial pres. | 4.0 atm         | 0   | 0   |
| Equili. pres. | $(4.0 - p)$ atm | $p$ | $p$ |

$$K_p = \frac{p_{\text{C}_2\text{H}_4} \cdot p_{\text{H}_2}}{p_{\text{C}_2\text{H}_6}} = \frac{p \cdot p}{4.0 - p}$$

$$0.04 = \frac{p^2}{4.0 - p}$$

$$\text{or } 0.16 - 0.04p = p^2$$

$$a = 1, b = 0.04, c = -0.16$$

We know that,

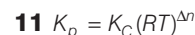
$$p = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$p = \frac{0.04 \pm \sqrt{0.0016 - 4(-0.16)}}{2}$$

$$p = \frac{-0.04 \pm 0.80}{2}$$

$$p = 0.38 \quad (\text{by taking positive value})$$

$$\text{Hence, } p_{\text{C}_2\text{H}_6} = 4.0 - 0.38 = 3.62 \text{ atm}$$



$$\text{Here, } \Delta n = 1$$

Thus,  $K_C$  will be less than  $K_p$ .

**12** Mathematical expression of the thermodynamic equilibrium is

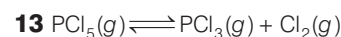
$$\Delta G = \Delta G^\circ + 2.303 RT \log Q$$

At equilibrium, when  $\Delta G = 0$

and  $Q = K$ , then

$$\Delta G = \Delta G^\circ + 2.303 RT \log K = 0$$

$$\Delta G^\circ = -2.303 RT \log K$$



$$\Delta n = 2 - 1 = 1$$

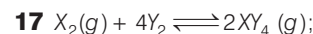
Value of  $\Delta n$  is positive, so the dissociation of  $\text{PCl}_5$  take forward by decrease in the pressure and by increase in the pressure the dissociation of  $\text{PCl}_5$  decreases.

**14**  $\Delta H$  is positive, so it will shift towards the products by increase in temperature.

**15** According to Le-Chatelier's principle, when concentration of reactant increases, the equilibrium shift in favour of forward reaction.

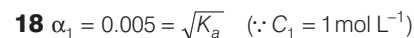
**16** According to Le-Chatelier's principle :

By increasing the pressure and decreasing temperature the equilibrium shifts in forward direction.



$$\Delta H < 0 \text{ and } \Delta n < 0$$

$\therefore$  The forward reaction is favoured at high pressure and low temperature. (According to Le-Chatelier's principle).



Molarity of the diluted solution,

$$C_2 = \frac{2}{32} = \frac{1}{16} \text{ mol L}^{-1}$$

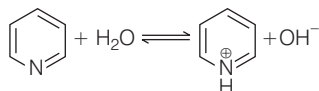
$$\therefore \alpha_2 = \frac{\sqrt{K_a}}{\sqrt{C_2}} = \frac{\sqrt{K_a}}{\sqrt{C_2}} \\ = 0.005 \times \sqrt{16} = 0.02$$

$$\text{Again } [\text{H}_3\text{O}^+] = C_2 \alpha_2$$

$$= \frac{1 \times 0.002}{16}$$

$$= 1.25 \times 10^{-3} \text{ M}$$

- 19** The percentage of pyridine can be equal to the percentage of dissociation of pyridinium ion and pyridine solution as shown below:

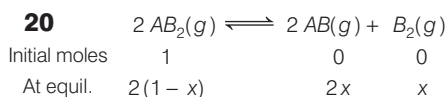


As pyridinium is a weak base, so degree of dissociation is given as :

$$\alpha = \sqrt{\frac{K_b}{C}} = \sqrt{\frac{1.7 \times 10^{-9}}{0.10}} = \sqrt{1.7 \times 10^{-8}} = 1.3 \times 10^{-4}$$

or, percentage of dissociation

$$= (\alpha \times 100)\% = (1.3 \times 10^{-4}) \times 100 = 0.013\%$$



where,  $x$  = degree of dissociation

Total moles at equilibrium

$$= 2 - 2x + 2x + x = (2 + x)$$

$$\text{So, } p_{\text{AB}_2} = \frac{2(1-x)p}{(2+x)}, \quad p_{\text{AB}} = \frac{2xp}{(2+x)}$$

$$p_{\text{B}_2} = \frac{xp}{(2+x)}$$

$$K_p = \frac{(p_{\text{AB}})^2 (p_{\text{B}_2})}{(p_{\text{AB}_2})^2}$$

$$= \frac{\left(\frac{2xp}{2+x}\right)^2 \left[\left(\frac{x}{2+x}\right)p\right]}{\left[\left(\frac{2(1-x)}{2+x}\right)p\right]^2}$$

$$= \frac{4x^3 p^3}{(2+x)^3} \times \frac{(2+x)^2}{p^2 4(1-x)^2}$$

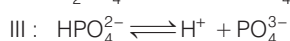
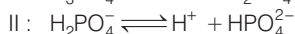
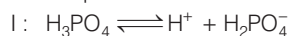
$$= \frac{x^3 p}{(2+x)(1-x)^2}$$

$$= \frac{x^2 p}{2} \quad [\because x \ll 1 \text{ and } 2]$$

$$x = \left(\frac{2K_p}{p}\right)^{1/3}$$

so,  $(1-x) \approx 1$  and  $(2+x) \approx 2$

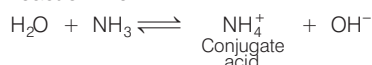
- 21**  $\text{H}_3\text{PO}_4$  is a tribasic acid, thus ionise in three steps.



Conjugate base is formed when an acid loses its proton. Thus,  $\text{HPO}_4^{2-}$  is the conjugate base of  $\text{H}_2\text{PO}_4^-$  (which is an acid in step II, but is the conjugate base of  $\text{H}_3\text{PO}_4$  in step I).

- 22**  $\text{BF}_3$  is an electron deficient species, thus behaves like a Lewis acid.

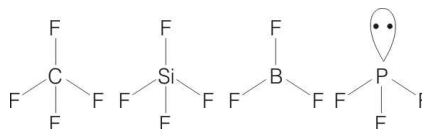
- 23** Ammonium ion is a conjugate acid. Water behaves both as an acid as well as a base. It is amphoteric solvent. In reaction like



It behaves like an acid.

- 24** The molecule with lone pair at centre atom, will behave as Lewis base.

In the given molecules, only  $\text{PF}_3$  has lone pair at P as shown below:



Thus,  $\text{PF}_3$  acts as a Lewis base (electron-pair donor) due to presence of lone pair on P-atom.

- 25** Boron is an element of 13 group and contains three electrons in its valence shell. When its compound  $\text{BH}_3$  dimerises, each boron atom carry only 6 electrons that is their octet is incomplete. Hence,  $(\text{BH}_3)_2$  is an electron deficient compound.

In all other given molecules, octet of central atom is complete.

$$\text{26 } K_{a1} = \frac{[\text{H}^+][\text{HS}^-]}{[\text{H}_2\text{S}]}$$

$$K_{a2} = \frac{[\text{H}^+][\text{S}^{2-}]}{[\text{HS}^-]}$$

$$K_{a3} = \frac{[\text{H}^+]^2 [\text{S}^{2-}]}{[\text{H}_2\text{S}]}$$

$$K_{a3} = K_{a1} \times K_{a2}$$

$$\begin{aligned} \text{27 } K &= K_{a1} \times K_{a2} \\ &= 4.5 \times 10^{-3} \times 1.7 \times 10^{-10} \\ &= 7.65 \times 10^{-13} \end{aligned}$$

$$\begin{aligned} [\text{H}^+] &= \sqrt{K_C} = \sqrt{7.65 \times 10^{-13} \times 0.01} \\ &= 8.7 \times 10^{-8} \end{aligned}$$

$$\text{pH} = -\log [\text{H}^+] = -\log 8.7 \times 10^{-8}$$

$$\begin{aligned} &= -(\log 10^{-8} + \log 8.7) = 8 - 0.93 \\ &= 7.07 \quad (\text{Answer is nearer to } d) \end{aligned}$$

- 28** pH of the solution A = 3

$$[\text{H}^+]_A = 10^{-3} \text{ M}$$

pH of the solution B = 2

$$[\text{H}^+]_B = 10^{-2} \text{ M}$$

$$[\text{H}^+] = 10^{-3} + 10^{-2} = 11 \times 10^{-3}$$

$$\begin{aligned} \text{pH} &= -\log (11 \times 10^{-3}) = 3 - \log 11 \\ &= 3 - 1.04 = 1.9 \end{aligned}$$

- 29** pH = 5.4,

$$\therefore [\text{H}^+] = 10^{-5.4} = 10^{-6} \cdot 10^{0.6}$$

Antilog of 0.6 is  $\approx 3.98$

$$[\text{H}^+] = 3.98 \times 10^{-6} \text{ M}$$

- 30** pH of  $\text{HOCl}$  = 2.85

But,  $-\text{pH} = \log [\text{H}^+]$

$$\therefore -2.85 = \log [\text{H}^+] \Rightarrow 3.15 = \log [\text{H}^+]$$

$$\Rightarrow [\text{H}^+] = 1.413 \times 10^{-3}$$

For weak monobasic acid,

$$\begin{aligned} [\text{H}^+] &= \sqrt{K_a \times C} \\ \Rightarrow K_a &= \frac{[\text{H}^+]^2}{C} = \frac{(1.413 \times 10^{-3})^2}{0.08} \\ &= 24.957 \times 10^{-6} \\ &= 2.4957 \times 10^{-5} \end{aligned}$$

- 31** 75 mL  $\frac{M}{5}$   $\text{HCl}$  + 25 mL  $\frac{M}{5}$   $\text{NaOH}$

$$\begin{aligned} \text{Milliequivalent of HCl} &= 75 \text{ mL of } \frac{M}{5} \text{ HCl} \\ &= \frac{1}{5} \times 75 = 15 \end{aligned}$$

Milliequivalent of  $\text{NaOH}$

$$= 25 \text{ mL of } \frac{M}{5} \text{ NaOH} = \frac{1}{5} \times 25 = 5$$

$\therefore$  Milliequivalent of  $\text{HCl}$  left unused

$$= 15 - 5 = 10$$

Volume of solution = 100 mL

$\therefore$  Molarity of  $[\text{H}^+]$  in the resulting mixture

$$= \frac{10}{100} = \frac{1}{10}$$

$$\text{pH} = \log \frac{1}{[\text{H}^+]} = \log(10) = 1$$

- 32** 20 mL of 0.1  $\text{NHCl}$  =  $\frac{0.1}{1000} \times 20 \text{ g eq}$   
 $= 2 \times 10^{-3} \text{ g eq}$

20 mL of 0.001 N  $\text{KOH}$

$$\begin{aligned} &= \frac{0.001}{1000} \times 20 \text{ g eq} \\ &= 2 \times 10^{-5} \text{ g eq} \end{aligned}$$

∴ HCl left unneutralised

$$= 2 (10^{-3} - 10^{-5})$$

$$= 2 \times 10^{-3} (1 - 0.01)$$

$$= 2 \times 0.99 \times 10^{-3}$$

$$= 1.98 \times 10^{-3} \text{ g eq}$$

Volume of solution = 40 mL

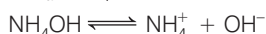
$$[\text{HCl}] = \frac{1.98 \times 10^{-3}}{40} \times 1000 \text{ M}$$

$$= 4.95 \times 10^{-2}$$

$$\text{pH} = 2 - \log 4.95 = 2 - 0.7 = 1.3$$

- 33**  $\text{BaCl}_2$  is a salt of strong acid HCl and strong base  $\text{Ba}(\text{OH})_2$ . So, its aqueous solution is neutral with pH 7. All other salts give acidic solution due to cationic hydrolysis, so their pH is less than 7. Thus, pH value is highest for the solution of  $\text{BaCl}_2$ .

- 34** Given,  $K_a (\text{NH}_4\text{OH}) = 1.77 \times 10^{-5}$



$$K_a = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} = 1.77 \times 10^{-5} \dots (i)$$

Hydrolysis of  $\text{NH}_4\text{Cl}$  takes place as,  
 $\text{NH}_4\text{Cl} + \text{H}_2\text{O} \longrightarrow \text{NH}_4\text{OH} + \text{HCl}$   
 or  $\text{NH}_4^+ + \text{H}_2\text{O} \longrightarrow \text{NH}_4\text{OH} + \text{H}^+$

Hydrolysis constant,

$$K_h = \frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} \dots (ii)$$

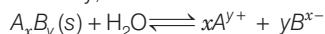
$$\text{or } K_h = \frac{[\text{NH}_4\text{OH}][\text{H}^+][\text{OH}^-]}{[\text{NH}_4^+][\text{OH}^-]} \dots (iii)$$

From Eqs. (i), (ii) and (iii)

$$K_h = \frac{K_w}{K_a} \quad [\because [\text{H}^+][\text{OH}^-] = K_w]$$

$$= \frac{10^{-14}}{1.77 \times 10^{-5}} = 5.65 \times 10^{-10}$$

- 35** For a sparingly soluble salt, if  $S$  is the molar solubility,

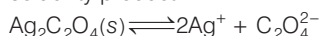


At saturation,

$$K[A_xB_y] = [A^{y+}]^x \times [B^{x-}]^y = [xS]^x [yS]^y$$

$$\text{or } K_{sp} = x^y \cdot y^y S^{x+y}$$

Where, the constant  $K_{sp}$  is called solubility product.



$$K_{sp} = [\text{Ag}^+]^2 [\text{C}_2\text{O}_4^{2-}] = [2S]^2 [S]$$

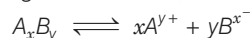
$$\text{Given, } 2S = 2.2 \times 10^{-4}$$

$$\text{or } S = 1.1 \times 10^{-4} \text{ M}$$

$$\therefore K_{sp} = [2.2 \times 10^{-4}]^2 [1.1 \times 10^{-4}]$$

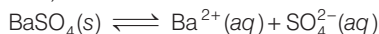
$$= 5.3 \times 10^{-12}$$

- 36** For a general reaction,



Solubility product ( $K_{sp} = [A^{y+}]^x [B^{x-}]^y$ )

For  $\text{BaSO}_4$  (binary solute giving two ions)



$$\therefore K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}] = (S)(S) = S^2$$

... (i) [where,  $S$  = solubility]

$$\text{Given, } S = 2.42 \times 10^{-3} \text{ g L}^{-1}$$

$$\text{Molar mass of } \text{BaSO}_4 = 233 \text{ g mol}^{-1}$$

∴ Solubility of  $\text{BaSO}_4$

$$(S) = \frac{2.42 \times 10^{-3}}{233} \text{ mol L}^{-1}$$

$$= 1.04 \times 10^{-5} \text{ mol L}^{-1}$$

On substituting the value of  $S$  in Eq. (i), we get

$$K_{sp} = (1.04 \times 10^{-5} \text{ mol L}^{-1})^2$$

$$= 1.08 \times 10^{-10} \text{ mol}^2 \text{ L}^{-2}$$

- 37** (a) For  $MY$ ,  $MY \rightleftharpoons M^+ + Y^-$

where,  $S$  = solubility

and  $K_{sp}$  = solubility product.

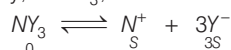
$$\therefore K_{sp} = [M^+][Y^-]$$

$$K_{sp} = S^2$$

$$S = \sqrt{K_{sp}} = \sqrt{6.2 \times 10^{-13}}$$

$$= 7.874 \times 10^{-7}$$

Similarly, for  $NY_3$ ,



$$\therefore K_{sp} = [N^+][Y^-]^3 = S \times (3S)^3$$

$$K_{sp} = 27S^4$$

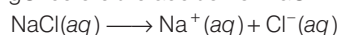
$$\therefore S = \sqrt[4]{\frac{K_{sp}}{27}} = \sqrt[4]{\frac{6.2 \times 10^{-13}}{27}}$$

$$= 3.89 \times 10^{-4}$$

Therefore, molar solubility of  $MY$  in water is less than that of  $NY_3$ .

- 38** As solubility of  $\text{AgCl}(s)$  is asked in 0.1 M NaCl solution, so in the calculation, solubility of  $\text{Cl}^-$  (from NaCl) must be added to the solubility of  $\text{Cl}^-$  (from AgCl).

Let,  $S$  be the solubility of  $\text{Ag}^+$  and  $\text{Cl}^-$  in AgCl before the addition of NaCl.



$$0.1 \text{ M} \quad 0 \quad 0$$

$$0 \quad 0.1 \text{ M} \quad 0.1 + S$$



$$S \quad S + 0.1$$

$$\text{Given, } K_{sp} = 1.6 \times 10^{-10} = [\text{Ag}^+][\text{Cl}^-]$$

$$\text{or } 1.6 \times 10^{-10} = S(0.1 + S) = 0.1S + S^2$$

∴  $K_{sp}$  is small, so  $S$  is very less in comparison with 0.1. Hence,  $S^2$  can be neglected.

$$\text{Thus, } 1.6 \times 10^{-10} = 0.1S$$

$$\text{or } S = 1.6 \times 10^{-9} \text{ M}$$

- 39** Given, pH of  $\text{Ba}(\text{OH})_2 = 12$

So,  $\text{pOH} = 2$

$$\therefore [\text{H}^+] = [1 \times 10^{-12}]$$

$$K_w = (\text{H}^+)(\text{OH}^-)$$

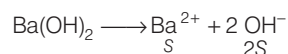
$$K_w = 1 \times 10^{-14}$$

$$\text{OH}^- = \frac{K_w}{\text{H}^+}$$

$$\text{and } [\text{OH}^-] = \frac{1 \times 10^{-14}}{1 \times 10^{-12}}$$

$$[\because [\text{H}^+][\text{OH}^-] = 1 \times 10^{-14}]$$

$$= 1 \times 10^{-2} \text{ mol/L}$$



$$K_{sp} = [\text{Ba}^{2+}][\text{OH}^-]^2 = [S][2S]^2$$

$$= \left[ \frac{1 \times 10^{-2}}{2} \right] (1 \times 10^{-2})^2$$

$$= 0.5 \times 10^{-6} = 5.0 \times 10^{-7} \text{ M}^3$$

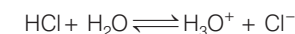
- 40** Among the given,  $\text{AgCl}$  is most soluble in aqueous ammonia solution, because it forms soluble complex with it.

- 41** A buffer solution is used in the removal of  $\text{PO}_4^{3-}$  ions.

- 42** NaCl is salt of strong acid and strong base. Its not the case of buffer.



Aqueous NaCl, itself exact neutral solution.



makes solution acidic.

- 43**  $\text{CH}_3\text{NH}_2 + \text{HCl} \longrightarrow \text{CH}_3\text{NH}_3\text{Cl}$

$$0.1 \quad 0.08 \quad 0$$

$$0.02 \quad 0 \quad 0.08$$

Basic buffer solution

$$\text{pOH} = \text{p}K_b + \log \frac{0.08}{0.02} = \text{p}K_b + 0.602$$

$$= 3.30 + 0.602 = 3.902$$

$$\text{pH} = 10.00$$

$$[\text{H}^+] = 7.99 \times 10^{-11} = 8 \times 10^{-11} \text{ M}$$

- 44** Strong acid with its salt cannot form buffer solution. Hence,  $\text{HClO}_4$  and  $\text{NaClO}_4$  is not an acidic buffer.



**45** By the use of Henderson's equation

$$pH = pK_a + \log_{10} \frac{[\text{salt}]}{[\text{acid}]}$$

When,  $[\text{salt}] = [\text{acid}]$

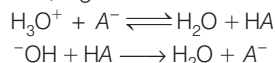
$$\therefore pH = pK_a$$

$\therefore pK_a = 3.58$ , thus at this state

$$pH = 3.58$$

So, acetoacetic acid ( $pK_a = 3.58$ ) is best to use.

**46** If small amount of an acid or alkali is added to a buffer solution, it converts them into unionised acid or base. Thus, its pH remains unaffected or in other words its acidity/alkalinity remains constant, e.g.



If acid is added, it reacts with  $A^-$  to form undissociated HA. Similarly, if base/alkali is added,  $OH^-$  combines with HA to give  $H_2O$  and  $A^-$  and thus, maintains the acidity/alkalinity of buffer solution.

**47**  $pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$

$$= -\log K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$= -\log 1.8 \times 10^{-5} + \log \frac{0.20}{0.30}$$

$$= 5 - 0.25 + (-0.176)$$

$$= 4.75 - 0.176 = 4.57$$

$$\therefore pH = 14 - 4.57 = 9.43$$

**48** (d)  $CH_3COOH$  (weak acid) and  $CH_3COONa$  (conjugated salt) form acidic buffer and for acidic buffer,

$$pH = pK_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

$$\text{and } [H^+] = -\text{antilog } pH$$

$$pH = -\log K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

$$[\therefore pK_a = -\log K_a]$$

$$= -\log (1.8 \times 10^{-5}) + \log \frac{(0.20)}{(0.10)}$$

$$= 4.74 + \log 2$$

$$= 4.74 + 0.3010 = 5.041$$

$$\text{Now, } [H^+] = \text{antilog } (-5.045)$$

$$= 9.0 \times 10^{-6} \text{ mol/L}$$

**49** (d) (i) For basic buffer,

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$(ii) pH + pOH = 14$$

$$\text{Given, } K_b = 1 \times 10^{-10}, [\text{salt}] = [\text{base}]$$

$$pOH = -\log K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$\therefore pOH = -\log (1 \times 10^{-10}) + \log 1 = 10$$

$$pH + pOH = 14$$

$$[\therefore \text{concentration of } [B^-] = [HB]]$$

$$pH = 14 - 10 = 4$$

**50** When we add  $NH_4OH$  in  $NH_4Cl$  solution, ionisation of  $NH_4OH$  is decreased due to common ion effect.

## SESSION 2

**1**  $pOH = -\log K_b + \log \left( \frac{\text{salt}}{\text{base}} \right)$

$$pOH = -\log [1.8 \times 10^{-5}] + \log \left[ \frac{\text{salt}}{1.0} \right]$$

$$5 = 4.7 + \log \left[ \frac{\text{salt}}{1.0} \right]$$

$$\log \left( \frac{\text{salt}}{1.0} \right) = 5 - 4.7 = 0.3$$

$$\left( \frac{\text{salt}}{1.0} \right) = \text{antilog } 0.3$$

$$\left( \frac{\text{salt}}{1.0} \right) = 1.995$$

$$[\text{salt}] = 1.99$$

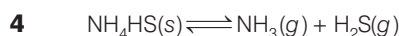
**2** (i)  $(K_p)_I = p_{CO_2} = 2$  (ii)  $(K_p)_{II} = \frac{p_{CO}^2}{p_{CO_2}} = 6$



$$(K_p)_{III} + p_{CO}^2 = (K_p)_I (K_p)_2 = 12$$

(Only gaseous species are taken in the expression of  $K_p$ ).

**3**  $K_p$  is independent of pressure and concentration.



$$\text{Initial at } a \quad 0.5 \text{ atm} \quad 0$$

$$\text{Equili. at } (a - x) \quad 0.5 + x \quad x$$

$$\text{Total pressure} = 0.5 + 2x = 0.84$$

$$x = 0.17$$

$$K_p = p_{NH_3} \cdot p_{H_2S} = 0.67 \times 0.17 = 0.1139$$

**5** Buffer capacity,  $\beta = \frac{dC_{HA}}{d_{pH}}$

where,  $dC_{HA}$

= moles of acid added per litre

$d_{pH}$  = change in pH

$$dC_{HA} = \frac{\text{moles of acetic acid}}{\text{volume}}$$

$$= \frac{0.12/60}{250/1000} = \frac{1}{125}$$

$$\therefore b = \frac{1/125}{0.02} = \frac{1}{2.5} = 0.4$$



$$\text{At } t = 0 \quad 1 \quad 1 \quad 0 \quad 0$$

$$\text{At equi. } (1-x) \quad (1-x) \quad x \quad x$$

$$x = 0.5$$

$$K_C = \frac{[C][D]}{[A][B]} = 1$$

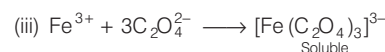
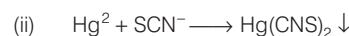


$$\begin{matrix} 1 & 2 & 0 & 0 \\ (1-y) & (2-y) & y & y \end{matrix}$$

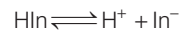
$$K_C = 1 = \frac{y^2}{(1-y)(2-y)}$$

$$y = 2/3$$

**7** The following reactions will disappear the red colour:



**8** Acid indicators are generally weak acid. The dissociation of indicator HIn takes place as follows :



$$K_{In} = \frac{[H^+][In^-]}{[HIn]} \quad [H^+] = K_{In} \cdot \frac{[HIn]}{[In^-]}$$

$$pH = -\log [H^+] = -\log \left( K_{In} \cdot \frac{[HIn]}{[In^-]} \right)$$

$$= -\log K_{In} + \log \frac{[In^-]}{[HIn]}$$

$$= pK_{In} + \log \frac{[In^-]}{[HIn]} = \log \frac{[In^-]}{[HIn]}$$

$$= pH - pK_{In}$$

**9** As the acidity or  $K_a$  value increases, pH decreases, thus the order of pH value of the acids is

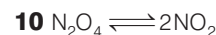
$$\text{Hypochlorous acid} > \text{acetic acid} >$$

$$(3.0 \times 10^{-8}) \quad (1.74 \times 10^{-5})$$

formic acid

$$(1.8 \times 10^{-4})$$

$$\left[ \text{Note } K_a \propto \frac{1}{pH} \right]$$



$$\begin{matrix} 1 & 0 \\ (1-x) & 2x \end{matrix}$$

Number of moles and volume

$$(\text{since, } pV = nRT \therefore V \propto n)$$

$$\text{Total moles} = (1+x)$$

$$\therefore \% \text{ of } NO_2 \text{ by volume}$$

$$= \frac{2x}{1+x} \times 100 = 50$$

$$x = \frac{1}{3} = 0.33$$

**11**  $K_p = K_c (RT)^{\Delta n}$

$$K_p = K_x (p)^{\Delta n}$$

$$\therefore K_c (RT)^{\Delta n} = K_x (p)^{\Delta n}$$

$$K_c = K_x \left( \frac{p}{RT} \right)^{\Delta n}$$

$$= K_x \left( \frac{1}{V} \right)^{\Delta n}$$

$$= K_x \left( \frac{1}{V} \right)^{-1}$$

$$= K_x (V)$$

**12**  $X^- + H_2O \rightleftharpoons HX + OH^-$

$$K_h = \frac{10^{-14}}{10^{-5}} = 10^{-9}$$

$$x = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{10^{-9}}{10^{-1}}} = 10^{-4}$$

$$100 \times 10^{-4} = 10^{-2} = 0.01\%$$

**13** On adding I, II and III and we get IV

$$\therefore K = \beta_1 \beta_2 \beta_3$$

$$\text{or } \log K = \log \beta_1 + \log \beta_2 + \log \beta_3$$

$$\text{or } -\log K = -\log \beta_1 - \log \beta_2 - \log \beta_3$$

$$p_K = p_{\beta_1} + p_{\beta_2} + p_{\beta_3}$$

$$\text{Where, } p_x = -\log x$$

**14**  $2NH_3 \rightleftharpoons N_2(g) + 3H_2(g)$

|         |        |   |    |
|---------|--------|---|----|
| Initial | a      | 0 | 0  |
| Equil.  | (a-2x) | x | 3x |

$$x = 0.30$$

$$a - 2x = 0.30$$

$$\therefore a = 0.90$$

Thus, initial moles of  $NH_3$  = 0.90 mol

$$K_c = \frac{[N_2][H_2]^3}{[NH_3]^2}$$

$$= \frac{\left( \frac{0.30}{0.50} \right) \left( \frac{0.90}{0.50} \right)^3}{\left( \frac{0.30}{0.50} \right)^2}$$

$$= 9.72$$

Thus,  $K_c$  for  $NH_3$  formation

$$= \frac{1}{9.72} = 0.103$$

**15** x mol  $(CH_3COOH)_2 \cdot Pb$

$$= 2x \text{ mol } CH_3COO^-$$

$$[CH_3COOH] = 0.1 \text{ M}$$

$$[CH_3COO^-] = 2x \text{ M}$$

$$\therefore pH = pK_a + \log \frac{[CH_3COO^-]}{[CH_3COOH]}$$

$$5.04 = 4.74 + \log 20x$$

$$0.30 = \log 20x$$

$$x = 0.1 \text{ mol}$$

**16** Which has least solubility, i.e.  $\sqrt{K_{sp}}$   
option c is correct, as its solubility is  $10^{-54}$  (least one).

**17** Equal pH means

$$[H^+]_1 = [H^+]_2$$

$$\sqrt{K_1 C_1} = \sqrt{K_2 C_2}$$

(by Ostwald's dilution law)

$$\text{or } K_1 C_1 = K_2 C_2$$

$$\therefore \frac{C_1}{C_2} = \frac{K_2}{K_1} = \frac{1}{4} = 0.25$$

**18** Salts  $(NH_4A, NH_4B, NH_4C)$  are of weak acid and weak base.

$$pH = 7 + \frac{pK_a}{2} - \frac{pK_b}{2}$$

Thus, greater the value of  $pK_a$  of HA, greater the pH.

$$pK_b(A^-, B^-, C^-) = \frac{A^-}{4} < \frac{B^-}{5} < \frac{C^-}{6}$$

$$pK_a(HA, HB, HC)$$

$$= HC < HB < HA$$

Greater the  $pK_a$

Greater the pH

Thus, pH

$$NH_4C < NH_4B < NH_4A$$

**19** I.  $Q = \frac{[NO_2]^2}{[N_2O_4]}$

Since,  $[NO_2]$  is maximum and  $[N_2O_4]$  is minimum at point A, hence Q is minimum at this point. Thus, false.

II.  $Q = \frac{(0.1)^2}{(0.1)} = (0.1) < K_c$

Thus, reaction proceeds-left to right.

Thus, true

III.  $K_c = Q$  when equilibrium is reached at point D or F. Thus, true.

**20**  $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

|         |       |       |       |       |
|---------|-------|-------|-------|-------|
| Initial | 1     | 1     | 1     | 1     |
| Equil.  | (1-x) | (1-x) | (1-x) | (1+x) |

$$K_c = \frac{[CO_2][H_2]}{[CO][H_2O]}$$

$$9 = \frac{(1+x)^2}{(1-x)^2}$$

$$\therefore \frac{1+x}{1-x} = 3$$

$$x = 0.5$$

$$\therefore [CO_2] = 1 + x = 1 + 0.5 = 1.5 \text{ mol}$$